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DATASET

SUPERVISED LEARNING FUNCTIONS!



In supervised learning, we have a dataset consisting of input-output pairs, usually denoted as $(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)$, where x is the input and y is the corresponding output. The goal is to learn a function f such that for a new input x_{new} , the function can predict the output y_{new} with accuracy ($y = f(x)$).

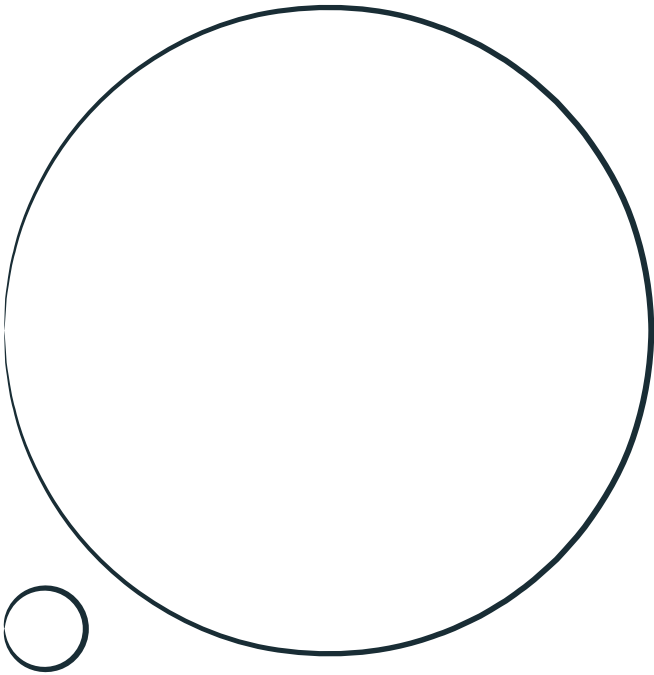
y

x

x'

y'

The idea of supervised learning is to learn a function f based on the training data and then test its performance on previously unseen data.



usually denoted as



such that for any

is the input and



, where

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with accuracy (

input

(x_1, y_1)

(x_2, y_2)

$\cdot \cdot \cdot (x_n, y_n)$







x_{new}

y_{new}

$$y = f(x)$$

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THE MAIN QUESTION: CAN WE LEARN!

